



PREDICTIVE MODELLING OF MATERNAL HEALTH RISK USING MACHINE LEARNING APPROACHES

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Predictive Modelling of Maternal Health Risk Using Machine Learning Approaches

V Mohammed Raiyyan

Abstract

Background: Maternal health is a critical area in healthcare, as complications during pregnancy can directly impact the well-being of both the mother and child. Early identification of risk factors plays a vital role in preventing adverse outcomes. In this study, the Maternal Health Risk Assessment Dataset (MHRAD) is used to build predictive models for classifying maternal health risk levels. This dataset comprises detailed clinical, physiological, and historical health information collected from maternal patients to evaluate potential health risks during pregnancy.

Objective: The main objective of this research is to develop and evaluate machine learning models that can accurately predict maternal health risk levels. By leveraging patient health data, the study seeks to support healthcare professionals in identifying high-risk cases early, thereby improving maternal care and reducing preventable complications.

Methods: Supervised Machine Learning models such as Logistic Regression, Decision Trees, Random Forest and Support Vector Machine were trained and tested on the dataset. The predictive performance of the models was evaluated using multiple metrics, including accuracy, precision, recall, and F1-score. To facilitate user interaction, a Graphical User Interface (GUI) was developed using the Tkinter library in Python.

Results: Logistic Regression achieved an accuracy of 97% with precision, recall and F1-scores of 99%, 96% and 97% respectively. The Random Forest model also attained an accuracy of 97%, with precision, recall and F1-scores of 100%, 96% and 89% respectively. The Decision Tree classifier performed comparatively well with 96% accuracy and balanced precision and recall values, while the Support vector classifier showed relatively lower performance compared to the other models.

Conclusion: This study demonstrates that machine learning can serve as an effective tool for predicting maternal health risks. By enabling early and reliable detection of high-risk cases, such approaches can empower clinicians to provide timely interventions.

Keywords: Maternal health risk, preeclampsia, sustainable development goal

Abbreviations: WHO – World Health Organization, MHRAD - Maternal Health Risk Assessment Dataset, SDG - Sustainable Development Goal, LR - Logistic Regression, RF - Random Forest, DTC - Decision Trees Classifier, SVC - Support Vector Classifier, EDA - Exploratory Data Analysis, GUI - Graphical User Interface.

1. Introduction

Maternal health remains one of the most significant challenges in global public health, as pregnancy and childbirth complications continue to claim the lives of hundreds of thousands of women annually. According to the World Health Organization (WHO), about 2,60,000 women died during and following pregnancy and childbirth in 2023, with 92% of these deaths occurred in low- and middle-income countries, particularly in sub-Saharan Africa and South Asia. More than 700 women died daily from

preventable pregnancy- and childbirth-related causes, averaging nearly one maternal death every two minutes [1]. These deaths are largely preventable, yet persist due to a combination of biological, social, and systemic factors. The leading causes of maternal mortality include obstetric haemorrhage (~27%), hypertensive disorders of pregnancy such as preeclampsia and eclampsia (~16%), pregnancy-related infections (~10–15%), and complications from unsafe abortion (4–13%), while approximately 23% of maternal deaths are attributable to indirect causes such as anaemia, malaria, HIV/AIDS, and pre-existing chronic conditions that worsen during pregnancy [2]. These figures highlight the urgency of improving maternal care and developing preventive strategies.

Machine Learning (ML), a subset of artificial intelligence, is a data-driven approach that enables computer systems to learn patterns from data and make predictions or decisions without being explicitly programmed. Unlike traditional statistical methods, which often rely on predefined assumptions, ML algorithms adapt to data complexity, making them powerful tools for analyzing large and heterogeneous datasets. Over the past decade, ML has emerged as a transformative technology across industries, with healthcare and medicine being among its most promising areas of application. Pregnancy risk prediction using modern machine learning approaches will be a promising strategy to address this burden by enabling the early identification of women at high risk for complications, thereby facilitating timely interventions, reducing preventable deaths, and optimizing limited healthcare resources. Predictive models can integrate routinely available health indicators such as age, blood pressure, heart rate, and body mass index to stratify maternal risk levels, which is particularly valuable in resource-constrained settings where prioritization of care is necessary. For this purpose, the Mendeley Maternal Health Risk Assessment Dataset serves as a crucial research resource [9]. This dataset contains structured records of maternal health parameters with corresponding risk classifications, enabling the training and evaluation of supervised machine learning models for risk prediction. Its features reflect variables that are both clinically relevant and easily measurable during routine antenatal care, enhancing the translational potential of predictive tools derived from it. By leveraging such datasets and applying robust analytical techniques, predictive modelling can contribute significantly to the global effort to reduce maternal mortality and improve maternal health outcomes, aligning with the Sustainable Development Goal (SDG 3) target of reducing the global maternal mortality ratio to less than 70 per 100,000 live births by 2030 [3]. Therefore, the study aims to harness machine learning methods to develop reliable predictive models for maternal health risk classification, providing a data-driven foundation for early diagnosis and preventive healthcare interventions.

2. Literature Survey

Recent research has increasingly explored the application of machine learning techniques for predicting preeclampsia (PE) and related maternal health risks. Montgomery-Csobán et al. (2024) developed the PIERS-ML model, a machine learning-based risk assessment tool, which demonstrated improved prediction of short-term maternal outcomes compared to traditional clinical models, though its focus on 48-hour outcomes highlights the need for broader prospective validation [4]. Similarly, Wu et al. (2025) employed deep learning and ensemble approaches to achieve high accuracy in early, non-invasive detection of preeclampsia, emphasizing the potential of AI in personalized obstetric care; however, their study called for larger-scale external validation and real-world feasibility assessments [5]. Li et al. (2024) applied logistic regression, random forest, and ensemble models to integrate multiple biomarkers, enhancing the accuracy of PE prediction, yet the high cost of biomarker analysis limits its clinical applicability [6]. In another study, Tiruneh et al. (2024) utilized random forest and gradient boosting algorithms on routine maternal health data, achieving acceptable accuracy for PE prediction despite issues of data variability and the absence of external validation [7]. Finally, Qiao et al. (2025) explored machine learning models incorporating blood and inflammatory markers, demonstrating excellent early

prediction performance across multiple centres, though prospective studies remain necessary for clinical translation [8]. Collectively, these studies underscore the promise of machine learning in advancing early detection of preeclampsia, while also revealing persistent challenges in model generalizability, clinical integration, and cost-effectiveness.

3. Methodology

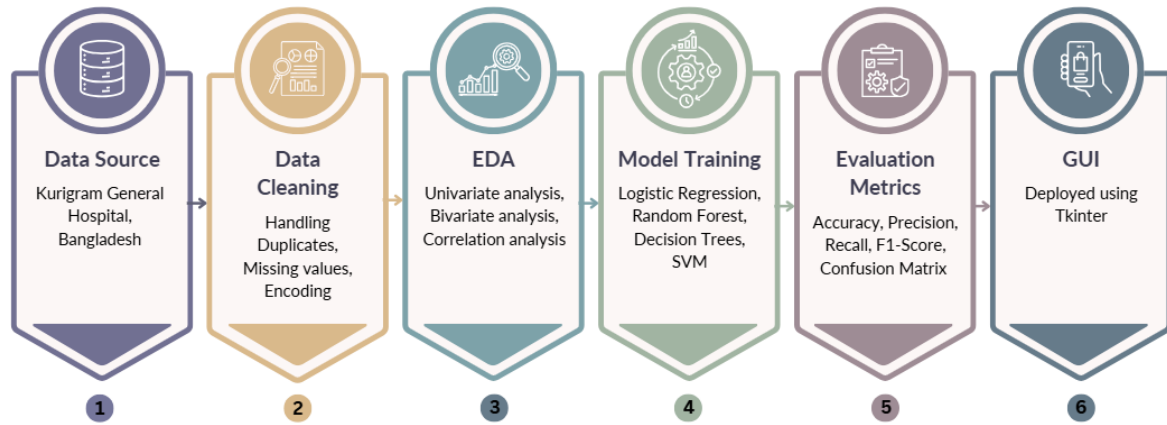


Figure 1 Research Flowchart

Figure 1 shows the workflow of the proposed system. The study utilized the publicly available dataset sourced online for analysis and model development. The workflow included data cleaning, exploratory data analysis, and training of machine learning models such as Logistic Regression, Random Forest, Decision Tree, and SVM, evaluated using standard performance metrics. Finally, a Tkinter-based GUI was developed to deploy the trained model for real-time maternal risk prediction in an accessible format.

3.1. Data Source

The dataset used in this study, published on Mendeley Data by Mojumdar et al. (2025), contains records originally collected from Kurigram General Hospital, Bangladesh. The dataset is multivariate and tabular, comprising *1,205 samples* which includes diverse range of physiological, demographics, medical and psychological attributes, along with one target variable. The predictor variables include Age (years), Systolic (mmHg), Diastolic Blood Pressure (mmHg), Blood Sugar (mmol/L), Body Temperature (°F), BMI (kg/m²), Previous Complications, Pre-existing Diabetes, Gestational Diabetes, Mental Health and Heart Rate (bpm). The response variable, Risk Level, is categorical with two classes - *Low* and *High*, representing levels of maternal health risk during pregnancy [9].

Characteristics	Type	Unit/Value	Mean ± SD
Age	Numeric	Years	27 ± 9.20
Systolic BP	Numeric	mmHg	116.82 ± 18.72
Diastolic	Numeric	mmHg	77.17 ± 14.31
BS	Numeric	mmol/L	7.5 ± 3.05
Body Temp	Numeric	°F	98.40 ± 1.09
BMI	Numeric	kg/m ²	23.32 ± 3.88
Previous Complications	Binary	0,1	No: 82%; Yes: 18%
Pre-existing Diabetes	Binary	0,1	No: 71%; Yes: 29%
Gestational Diabetes	Binary	0,1	No: 88%; Yes: 12%
Mental Health	Binary	0,1	No: 67%; Yes: 33%
Heart Rate	Numeric	bpm	75.82 ± 7.23

Table 1. Baseline Characteristics

3.2. Data Cleaning

Data preprocessing was performed to ensure the integrity and reproducibility of subsequent modelling. The raw dataset was first inspected for structural consistency, missingness, duplicates, invalid entries and anomalous values using summary statistics and graphical diagnostics. Redundant records, which may arise from repeated data entry or overlapping patient observations, were identified using exact row duplication detection and subsequently removed. This procedure was essential to prevent bias in model training and ensure that each observation contributed equally to the learning process. Missingness within the dataset was evaluated column-wise to determine its nature and extent. Continuous variables (e.g., systolic BP, diastolic bp, body sugar, BMI, and heart rate) exhibiting missing entries were imputed using the mean value of the corresponding attribute to mitigate the influence of extreme observations. Categorical or binary variables (e.g., Preexisting Diabetes, Previous Complications) were imputed using the mode, representing the most frequent category ^[10]. Inconsistent data might lead to erroneous conclusions and forecasts ^[11]. To enhance numerical stability and ensure comparability across variables measured on different scales, all continuous features were standardized using z-score normalization ^[10]. This procedure was particularly important for algorithms sensitive to feature magnitude, such as Logistic Regression and Support Vector Machines (SVM). The target attribute, Risk Level, originally expressed as categorical classes (*Low*, *High*), was encoded into numeric form (*0*, *1*) through label encoding. All preprocessing scripts were implemented and version-controlled in Google Colab.

3.3. Exploratory Data Analysis

3.3.1. Univariate Analysis

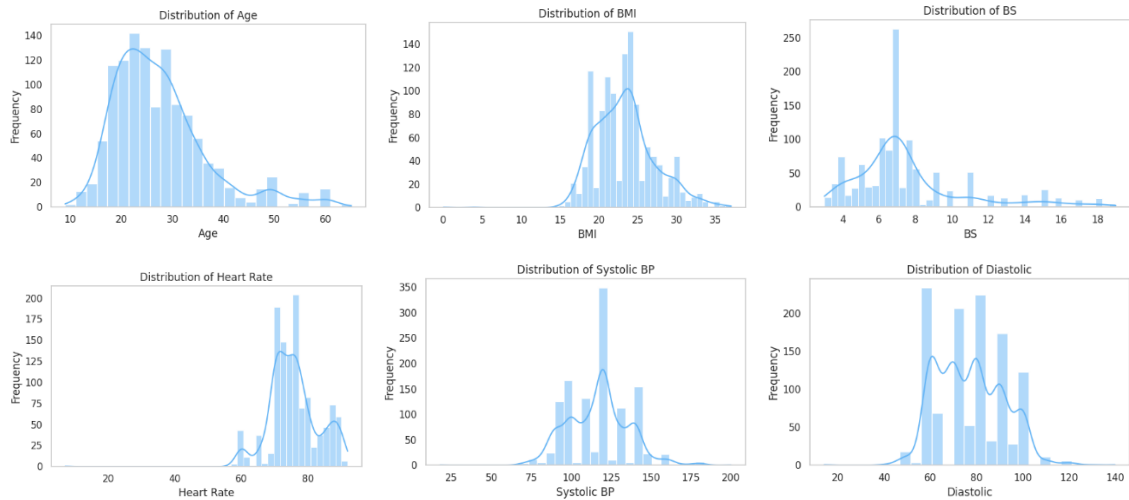


Figure 2 Frequency Distribution of Continuous Variables

Figure 2 illustrates the frequency distribution of major continuous clinical variables in the dataset, including age, BMI, blood sugar (BS), heart rate, systolic blood pressure, and diastolic BP. Each histogram is overlaid with a kernel density estimate to provide a visual assessment of distribution modality and spread. Most variables exhibit unimodal, approximately normal or mildly skewed distributions, except for BS, which displays positive skewness consistent with the presence of outliers or a larger pre-diabetic/diabetic subpopulation. These distributions provide a foundation for subsequent comparative analysis and highlight potential non-normality in critical predictors, informing later choices in statistical modelling and hypothesis testing.

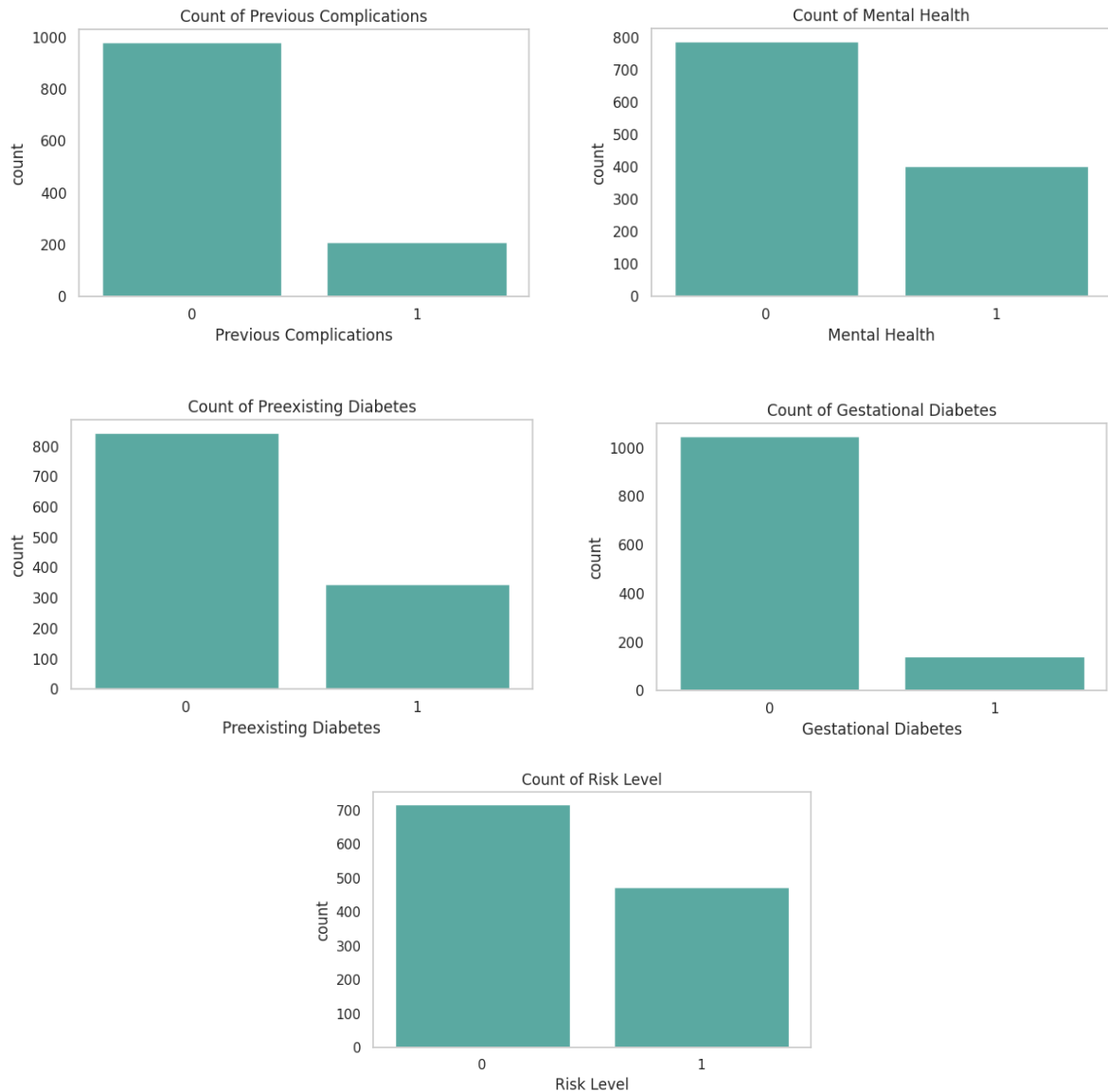


Figure 3 Distribution of Categorical Variables

Figure 3 presents the distribution of key categorical variables: previous complications, mental health issues, preexisting diabetes, gestational diabetes, and the risk level itself. The bar plots reveal that most participants had no previous complications, no mental health history, no gestational diabetes, and did not belong to the high-risk group. However, there remains substantial representation in all risk and disease categories, facilitating meaningful stratified analysis.

3.3.2. Bivariate Analysis

Figure 4 displays stratified bar plots of categorical variable prevalence by risk level. There are clear increases in the proportion of individuals with previous complications, mental health concerns, preexisting diabetes, and gestational diabetes among the high-risk group (Risk Level 1) compared to the low-risk group (Risk Level 0). This visualization underscores the associative trends between these comorbidities and elevated clinical risk status, substantiating their relevance as risk indicators and candidate predictors in further inferential modelling.

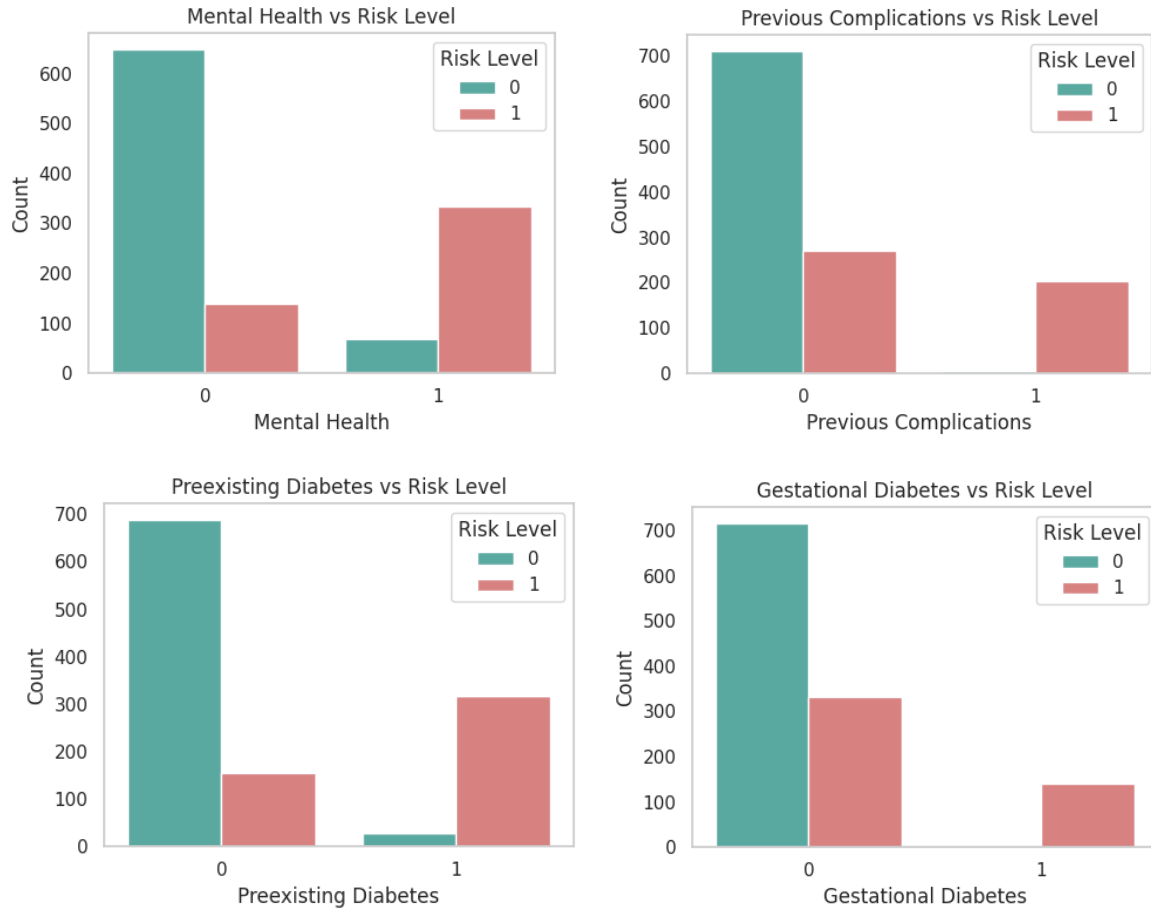
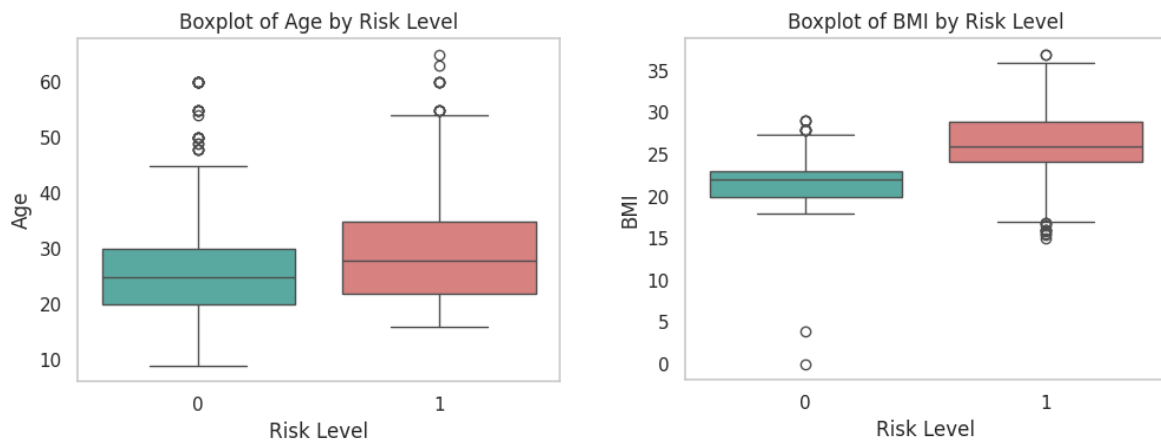


Figure 4 Categorical Variables by Risk Levels

Figure 5 exhibits comparative boxplots for all continuous variables, stratified by risk level. Higher median values of age, BMI, BS, heart rate, systolic BP, and diastolic BP are observed in the high-risk group, with some variables (e.g., BS and heart rate) demonstrating wider interquartile ranges and more outliers among high-risk participants. These findings indicate a general trend of elevated risk group individuals presenting with worse clinical and hemodynamic profiles, supporting their utility for risk stratification and guiding model development.



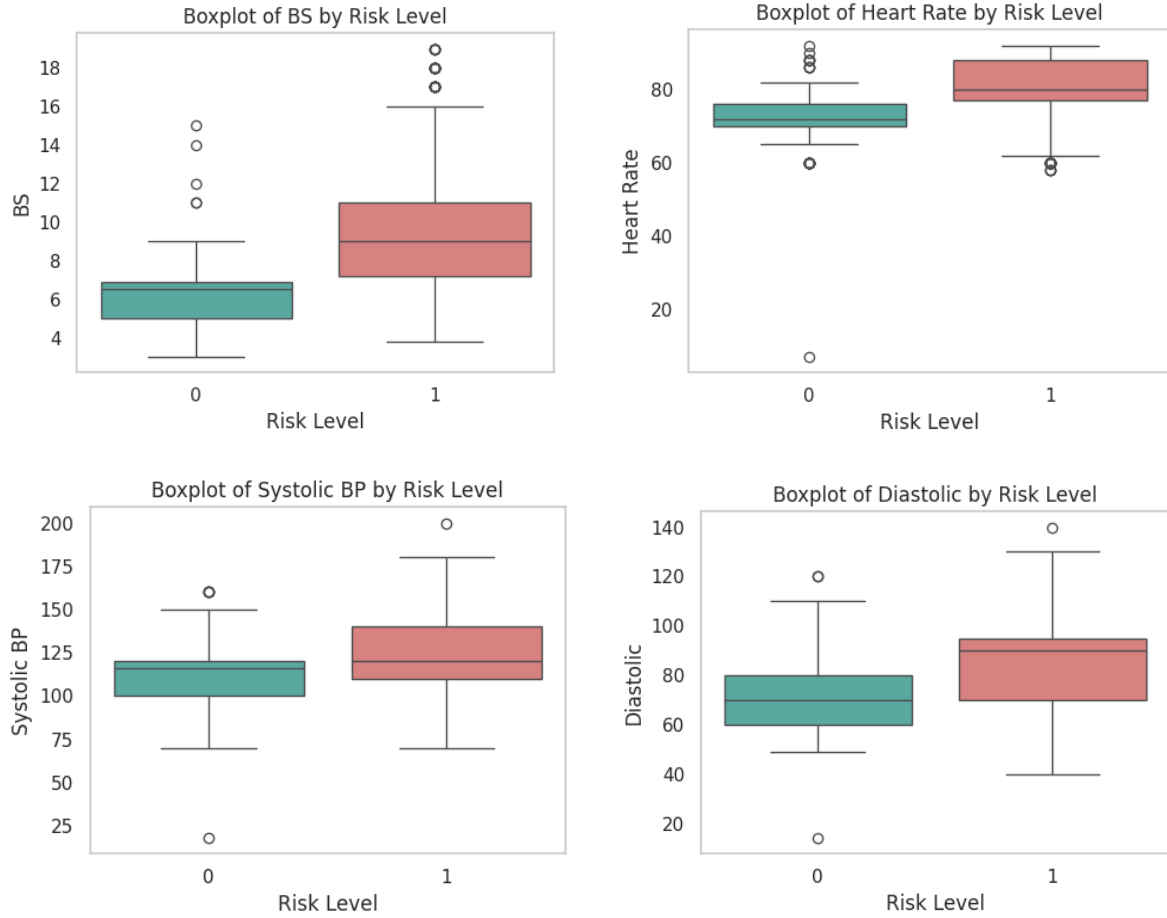


Figure 5 Continuous Variables by Risk Levels

3.4. Model Development

The study initially implemented **Logistic Regression (LR)**, **Decision Tree (DT)**, **Random Forest (RF)**, and **Support Vector Machine (SVM)** classifiers as baseline models.

3.4.1. Logistic Regression (LR)

Logistic Regression (LR) is a *supervised statistical learning algorithm* commonly used for *binary classification problems*, where the outcome variable takes two possible values (e.g., *High risk* vs *Low risk*). Unlike linear regression, which predicts a continuous output, Logistic Regression models the probability of a categorical dependent variable by mapping a linear combination of input features through a logistic (sigmoid) function [12]. It provides probabilistic outputs, enabling researchers to quantify uncertainty in predictions. The model's coefficients represent the log-odds of the dependent variable, allowing straightforward clinical interpretation — for instance, estimating how an increase in systolic blood pressure affects the probability of maternal health risk [13]. Its significance lies in its simplicity, transparency, and robustness, making it particularly suitable for healthcare studies where explainability and statistical inference are as important as predictive performance [14]. In this study, Logistic Regression was employed to predict risk levels using all available clinical and physiological attributes in the MHRAD — excluding the target variable, *Risk Level*.

3.4.2. Random Forest (RF)

Random Forest (RF) is an *ensemble-based supervised learning algorithm* introduced by Breiman (2001) that combines the predictive strength of multiple decision trees to improve accuracy and reduce overfitting [18]. It is widely recognized for its robustness, scalability, and ability to handle both numerical

and categorical data. A Random Forest constructs a large collection of decorrelated decision trees and aggregates their outputs to produce a final prediction. Each tree in the forest is trained on a different bootstrap sample of the training dataset, and at each node, a random subset of features is selected for splitting. This **randomization** enhances model diversity and significantly reduces the variance compared to a single decision tree [19]. The significance of RF lies in its capacity to model complex, non-linear relationships and to manage missing values, noise, and multicollinearity effectively. Moreover, it provides feature importance measures, allowing researchers to interpret which variables most influence the prediction of maternal risk outcomes—an essential consideration in clinical and epidemiological contexts [20].

3.4.3. Decision Tree (DT)

The Decision Tree (DT) algorithm is a *supervised learning model* that uses a tree-like structure of decisions and their possible consequences, including chance event outcomes and resource costs. It is one of the most interpretable machine learning algorithms, making it highly suitable for healthcare-related predictive modelling where transparency and explainability are essential [21]. A Decision Tree recursively partitions the input space into smaller, homogeneous subsets based on feature values, forming a hierarchy of decision nodes and leaf nodes. Each internal node represents a feature-based test, and each leaf node corresponds to a predicted class label [22]. The model operates on the principle of information gain or impurity reduction, seeking to minimize uncertainty in classification outcomes. The major significance of DTs lies in their interpretability, non-parametric nature, and ability to handle both categorical and continuous variables without extensive data preprocessing [23]. In this study, the Decision Tree classifier was utilized to predict maternal health risk levels based on all clinical indicators derived from the MHRAD dataset.

3.4.4. Support Vector Machine (SVM)

Support Vector Machine (SVM) is a *supervised machine learning algorithm* widely used for *classification and regression tasks*. It is particularly effective in high-dimensional spaces and for problems with a clear margin of separation between classes [15]. The primary objective of SVM is to find an optimal decision boundary, known as the **hyperplane**, that best separates the data points of different classes while maximizing the margin between them [16]. The margin represents the distance between the hyperplane and the closest data points from each class, known as **support vectors**. Maximizing this margin enhances generalization, allowing the model to perform well on unseen data. SVM is particularly significant in medical and healthcare applications because it can effectively model complex, non-linear patterns while maintaining interpretability through support vectors and kernel transformations [17]. Its strong mathematical foundation makes it a reliable choice when precision and stability are essential, such as in disease diagnosis or patient risk assessment. In this study, the SVM classifier was employed to predict risk levels based on physiological and clinical attributes. SVM was chosen due to its robustness against overfitting, ability to handle non-linear relationships through kernel functions, and strong theoretical foundations in statistical learning theory. The RBF kernel was used due to its flexibility and strong performance on non-linear data distributions.

3.5. Evaluation Metrics

3.5.1. Confusion Matrix

A **confusion matrix** is a tabular representation that illustrates the performance of a classification model by comparing the predicted labels against the actual labels. It consists of four key components:

Term	Description
True Positive (TP)	Number of correctly predicted <i>High-risk</i> cases.
True Negative (TN)	Number of correctly predicted <i>Low-risk</i> cases.
False Positive (FP)	<i>Low-risk</i> cases incorrectly classified as <i>High-risk</i> .
False Negative (FN)	<i>High-risk</i> cases incorrectly classified as <i>Low-risk</i> .

Table 2. Confusion Matrix

It provides a detailed breakdown of how well the model distinguishes between the two classes and helps identify types of misclassification errors [24][25].

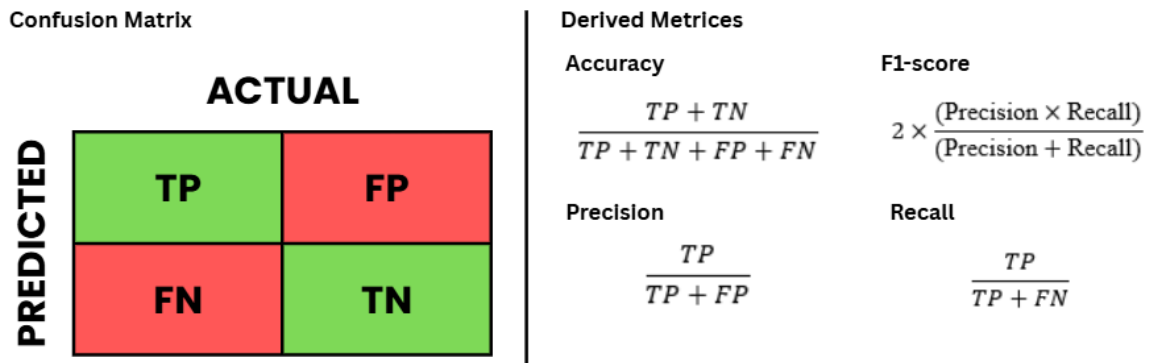


Figure 6 Evaluation Metrics

3.5.2 Derived Metrics

Accuracy: Accuracy measures the proportion of total correct predictions made by the model. It provides a general idea of overall performance but can be misleading for imbalanced datasets.

Precision: Precision (also known as Positive Predictive Value) indicates how many of the cases predicted as *High-risk* were actually *High-risk*. It reflects the model's ability to avoid false alarms.

Recall: Recall, or Sensitivity, measures how effectively the model identifies actual *High-risk* cases. A high recall means fewer *High-risk* patients are missed.

F1-Score: The **F1-score** is the harmonic mean of precision and recall. It balances the trade-off between the two, especially useful when class distribution is uneven or when both false positives and false negatives are critical [26].

3.6 GUI

To facilitate user interaction and make the predictive system accessible to non-technical users, a Graphical User Interface (GUI) was developed using the Tkinter library in Python. Tkinter, the standard GUI toolkit for Python, provides a lightweight yet powerful framework for building desktop-based interfaces. The GUI was designed to enable healthcare practitioners to input patient-specific clinical parameters—such as systolic and diastolic blood pressure, body mass index (BMI), blood sugar, and heart rate—and obtain an instant prediction of the maternal health risk level. The interface seamlessly integrates the trained and serialized Random Forest model, which was stored as a pickle file (.pkl) to ensure reusability and consistent performance without re-training. Upon data entry, the GUI automatically scales the inputs using the same preprocessing pipeline applied during model training and then generates the prediction in real-time. Clear text output and color-coded risk indicators were

incorporated to enhance interpretability and usability. This deployment demonstrates how a machine learning model can be transformed into a practical decision-support tool, providing an intuitive and interactive platform for early maternal risk assessment in clinical or field settings.

4. Results

The performance of four supervised machine learning models - Logistic Regression, Random Forest, Decision Tree and Support Vector Classifier was evaluated using standard performance metrics, including accuracy, precision, recall, and F1-score. Table 2 summarizes the comparative results obtained after training and testing on the MHRAD.

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.97	1.00	0.96	0.98
Logistic Regression	0.97	0.99	0.96	0.97
Decision Tree Classifier	0.96	0.98	0.96	0.97
Support Vector Classifier	0.85	0.83	0.96	0.89

Table 3. Comparison of Model Performance Metrics

Among the evaluated models, Random Forest demonstrated the highest overall performance, achieving an accuracy of 0.97 and an F1-score of 0.98, closely followed by Logistic Regression and Decision Tree Classifier, both of which also showed strong and consistent results. The Support Vector Classifier, while achieving a relatively lower accuracy of 0.85, exhibited a high recall value of 0.96, indicating its strength in identifying positive cases correctly, albeit at the cost of precision.

These results indicate that ensemble-based models, such as Random Forest, tend to perform more robustly on this dataset, likely due to their ability to handle non-linear feature interactions and mitigate overfitting compared to single classifiers. Owing to its superior predictive capability, robustness to overfitting, and interpretability through feature importance analysis, the Random Forest classifier was selected as the final model and integrated into a Tkinter-based graphical user interface (GUI). This GUI enables real-time maternal health risk prediction by allowing users to input clinical parameters and obtain immediate risk classifications, thus bridging the gap between machine learning research and clinical application.

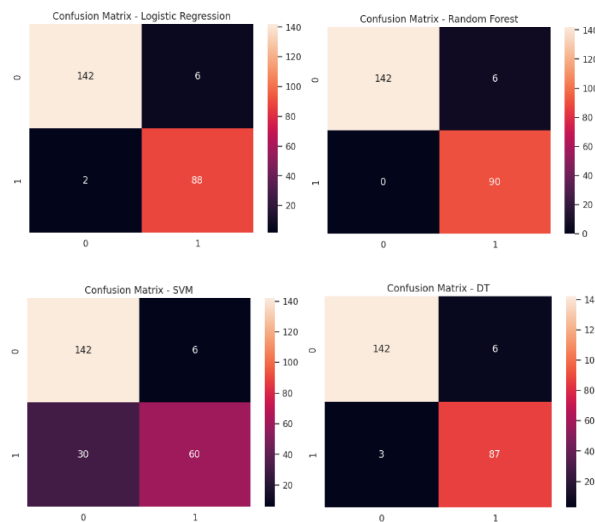


Figure 7 Confusion Matrices for Machine Learning Models

Pregnancy Risk Predictor

Age

26

Systolic BP

120

Diastolic

80

BS

7.1

Body Temp

98

BMI

21

Previous Complications

0 - No

Preexisting Diabetes

0 - No

Gestational Diabetes

0 - No

Mental Health

0 - No

Heart Rate

80

Predicted Risk Level: ☒ Low Risk

Predict Risk Level

Close

Pregnancy Risk Predictor

Age

35

Systolic BP

130

Diastolic

90

BS

8.4

Body Temp

98

BMI

24

Previous Complications

0 - No

Preexisting Diabetes

1 - Yes

Gestational Diabetes

1 - Yes

Mental Health

0 - No

Heart Rate

98

Predicted Risk Level: ☐ High Risk

Predict Risk Level

Close

Figure 8 Graphical User Interface

5. Discussion and Conclusion

The proposed model demonstrates the potential to serve as a decision-support tool for healthcare providers, particularly in resource-limited settings. By offering early and reliable risk assessments, it can assist in prioritizing high-risk pregnancies, guiding timely medical intervention, and ultimately reducing maternal morbidity and mortality. The GUI integration further enhances accessibility, allowing even non-technical users to interact with the predictive model efficiently. The findings reaffirm the effectiveness of machine learning in maternal health risk prediction, with ensemble models like Random Forest achieving superior performance compared to traditional methods. Logistic Regression and Decision Trees also performed competitively, validating the relevance of both linear and tree-based algorithms in structured clinical datasets. Importantly, the consistent accuracy across multiple models strengthens confidence in the dataset's predictive features and the overall methodological framework. Future research could focus on expanding the dataset with multi-institutional or longitudinal data to improve generalizability and external validation. Incorporating additional variables such as nutritional status, socio-demographic factors, and medical history may further refine prediction accuracy. Moreover, deploying the GUI as a web or mobile application and integrating it with hospital information systems could transform it into a real-time clinical decision support tool, enhancing its practical utility in maternal health monitoring.

Data Availability

Maternal Health Risk Assessment Dataset (Original data) ([Mendeley Data](#)).

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